

Evaluating Depth-Enhancement Shaders for Cerebral Vasculature in Passthrough Mixed Reality: A Standalone Mesh-Rendering System and Pre-Registered Protocol on Meta Quest 3

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Abstract. Accurately judging the depth of cerebral vessels is central to neurosurgical planning and image-guided navigation. We study the visualisation of cerebral angiography in video see-through (passthrough) Mixed Reality (MR) on a standalone Meta Quest 3. Full direct volume rendering of a brain volume proved too costly to sustain untethered while compositing over passthrough, so we render a segmented surface *mesh* of the prominent vasculature, which runs comfortably on-device and yields a cleaner, more interpretable stimulus. The open question we target is whether depth-enhancement shaders improve depth judgement when virtual anatomy is viewed against a real background — a regime between 2D displays, where such shaders clearly help, and immersive VR, where strong stereo and parallax make them wash out. We present a standalone MR system that registers the vessel mesh to the real workspace and a rigorously controlled, pre-registered protocol comparing three shaders — Phong (reference), aerial-perspective fog, and pseudo-chromadepth — on an absolute in-scene target-placement task. The design holds the base surface material and real-world occlusion constant across conditions, derives ground truth from the vessel centreline, and decomposes placement error into view-axis (depth) and lateral components. We report on-device rendering feasibility and discuss how the protocol is powered against the depth challenges specific to passthrough MR. **[TODO: add one-sentence headline result once the feasibility pilot is run.]**

Keywords: Mixed reality · Passthrough video see-through · Depth perception · Cerebral angiography · Surface rendering · Neurosurgical planning.

1 Introduction

Neurovascular interventions — including the planning and clipping of aneurysms, the resection of arteriovenous malformations (AVMs), and the management of arteriovenous fistulae (AVFs) — depend on a precise, three-dimensional understanding of how target vessels are situated within the surrounding brain [10]. A central difficulty is judging the *depth* of a vessel of interest: it is typically

embedded within, and partially occluded by, other anatomy, so its position relative to neighbouring vessels is hard to recover from the two-dimensional slices or flat desktop renderings used in conventional review, which provide weak and often ambiguous depth information [10]. **[TODO: one sentence on the clinical consequence of misjudging vessel depth, tied to collaborators’ use case if possible.]**

Video see-through (passthrough) Mixed Reality (MR) is an appealing medium for this task: a head-mounted stereoscopic display supplies binocular disparity and head-coupled motion parallax — two of the strongest depth cues — while registering the vasculature to the clinician’s real workspace keeps the physical environment as context. Realising this on a *standalone* headset, however, constrains the rendering. Raymarched direct volume rendering (DVR) of a brain volume is bandwidth- and fragment-shader-bound, and interactive medical DVR has been sustained only on PC-tethered or externally accelerated systems [5]; in our own testing it could not hold interactive framerates untethered while also compositing over the passthrough feed. We therefore render a segmented surface *mesh* of the prominent cerebral vasculature. The mesh runs comfortably within the standalone frame budget, and — because it dispenses with the transfer function whose opacity ramp otherwise entangles occlusion and aerial perspective with the manipulation of interest — yields a cleaner, more interpretable stimulus for a depth-perception study.

The open question is whether *depth-enhancement shaders* improve vessel depth judgement in this setting. On 2D displays, cues such as aerial perspective (fog) and pseudo-chromadepth substantially improve depth judgements of vascular data [8, 14]. In immersive VR, by contrast, binocular stereo and motion parallax are so strong that shader differences largely wash out [6, 17]. Passthrough MR sits between these regimes and adds its own difficulties: the virtual mesh competes with a real, textured background, and the passthrough imagery is geometrically distorted, reduced in resolution, and latent — the depth and occlusion problems long recognised in medical AR [13, 1]. Whether depth-enhancement shaders help angiographic depth judgement here is therefore open, and clinically consequential.

This paper makes three contributions:

- A *standalone* (untethered) passthrough-MR system that registers a cerebral-vasculature mesh to the real workspace on the Meta Quest 3 and renders it, with real-world occlusion, within the standalone frame budget (Section 3).
- A rigorously controlled, pre-registered *protocol* comparing three depth-enhancement shaders — Phong, fog, and pseudo-chromadepth — on an absolute in-scene target-placement task, with explicit confound controls and an error decomposition isolating the diagnostic depth axis (Section 4).
- An on-device *feasibility* check confirming that mesh rendering sustains headset framerate over the passthrough pipeline (Section 5).

2 Related Work

Depth cues for vascular visualisation. For rendered vasculature, local (Phong) surface shading is the conventional illumination baseline; on a surface mesh its normals are well defined, so it is a fair reference rather than a source of gradient noise. Added depth cues improve on it. Aerial perspective (fog) encodes distance as desaturation/attenuation, and pseudo-chromadepth — a depth-to-colour (red $\rightarrow$ blue) mapping introduced for angiography [14] — is consistently rated among the strongest static cues for vascular data [8]; it has been adapted specifically to vascular *surface* models [3], the setting we adopt. These results motivate our three-condition set: Phong (reference), fog, and pseudo-chromadepth. Depth-cue comparisons are well developed for surface renderings on conventional and VR displays [7], but have not been transferred to registered, passthrough-MR viewing of cerebral vasculature.

Depth perception in immersive VR and the ceiling effect. Much of that evidence comes from 2D displays. In stereoscopic VR with free head movement, disparity and motion parallax are themselves powerful cues, and several studies report that shader choice then matters little: depth judgements of vascular models were more precise in VR than on desktop yet barely distinguished shading methods [6], interactive cues had little impact on relative-depth perception for angiographic volumes in VR [17], and, once camera motion is permitted, depth perception depends more on scene content than on the chosen technique [12]. This ceiling effect motivates our move to passthrough MR and requires that difficulty be raised deliberately. Methodologically, 3D-pointing studies show that error along the view (depth) axis behaves differently from lateral error and is the axis on which depth cues act [16, 2], motivating an error decomposition rather than a single 3D distance.

Depth perception in passthrough augmented and mixed reality. Passthrough viewing changes the depth problem rather than easing it. The depth of virtual content overlaid on a real scene is systematically misperceived, modulated by display type, shadows, and relative position [1] and by object shape, colour, and luminance [4]; without correct occlusion, virtual structures appear to float in front of real ones, a long-standing, only partially solved problem [13, 15]. These effects are acute for medical overlays, and depth-enhancing vascular visualisation has been pursued specifically for neurovascular AR [9, 11]. The closest prior work compares depth cues for vasculature but on desktop or tethered VR and with relative-depth or detection tasks [8, 17]; we extend that comparison to *standalone passthrough MR* and to an *absolute* in-scene target-placement task whose error is decomposed into depth and lateral components, with the mesh registered to the real workspace and real-world occlusion enabled but held constant across conditions.

3 System

3.1 Vessel Mesh and Base Material

The stimulus is a segmented triangle *mesh* of the prominent cerebral vasculature, rendered as an opaque surface; the same mesh, skeletonised to its *centreline*, provides the ground truth used in the task (Section 4.1). Choosing a mesh over full DVR is what makes the system run untethered (Section 3.4) and removes the transfer function as a confound: with no opacity ramp, occlusion is simply correct surface occlusion and is held constant across conditions. A single base material — surface colour and Phong lighting response — is shared by all conditions, so each condition adds only its distance encoding on top of an identical baseline. **[TODO: dataset — modality, source, and number of cases — to be supplied; state mesh extraction (segmentation + surface reconstruction) once fixed.]**

3.2 Depth-Enhancement Shaders

The three conditions share the base material and differ only in the added depth cue: *Phong*, standard local surface shading, which serves as the reference; *fog*, an explicit parameterised depth→desaturation applied post-shading and disabled in the other conditions; and *pseudo-chromadepth*, a depth-dependent red→blue colour mapping [14, 3].

3.3 Passthrough Compositing and Registration

The system is built in Unreal Engine 5.7.3. The mesh is drawn over the Quest 3 passthrough feed using Meta’s Passthrough API and world-locked to the workspace with a Meta Spatial Anchor on the participant’s table, so it stays fixed as the participant moves. Real-world occlusion is resolved with the Quest 3 Depth API, so the participant’s hands and nearby objects correctly occlude the mesh; this occlusion is enabled and held constant across all conditions (Section 4.2).

3.4 Standalone Rendering on Quest 3

Unlike raymarched DVR, an opaque surface mesh is inexpensive to rasterise, so the system comfortably sustains the headset display rate alongside the passthrough and depth pipelines rather than the reduced target a standalone volume ray-march would impose. Measured performance is reported in Section 5.

4 Study Design and Protocol

4.1 Task and Ground Truth

The mesh is world-locked to the workspace with a Meta Spatial Anchor and viewed through the passthrough cameras, appearing as a solid object on the

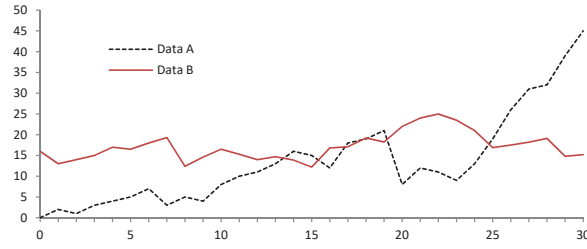


Fig. 1. The standalone cerebral-vasculature mesh composited over Quest 3 passthrough and world-locked with a Meta Spatial Anchor (in-headset capture). [TODO: replace placeholder.]

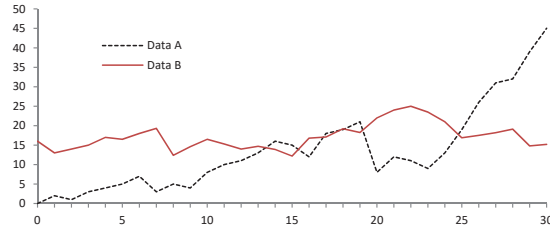


Fig. 2. The three depth-enhancement shaders applied to the *same* base material and geometry: (a) Phong (reference), (b) fog, (c) pseudo-chromadepth. [TODO: replace placeholder.]

participant’s table. On each trial the participant views the vasculature under one shader condition and indicates the perceived 3D location of a designated target vessel by positioning a virtual marker with a hand controller. Ground truth is the vessel *centreline* node nearest the intended landmark, obtained by skeletonising the segmented mesh; this gives an unambiguous anatomical point even though the rendered vessel has a finite radius. [TODO: confirm input modality (controller ray vs. direct grab) and marker form (point vs. small sphere).]

4.2 Conditions and Confound Controls

The single manipulated factor is the depth-enhancement shader, applied within subjects: Phong, fog, and pseudo-chromadepth (Section 3.2). The overriding control is that a *single base material* — surface colour and Phong response — and correct opaque occlusion are held constant across all conditions, so each shader changes only how distance is encoded, not the underlying surface or its occlusion [8, 17]. Fog is an explicit, parameterised depth-to-desaturation function *disabled* in the non-fog conditions, so the fog condition isolates an added distance encoding; in passthrough it blends toward a fixed reference colour rather than the variable real background, so it is not confounded with scene content. Stereo separation, motion parallax, the passthrough path, and real-world occlusion are

held identical across conditions; manipulating occlusion handling is left to future work (Section 6).

4.3 Procedure and Difficulty Calibration

After a short training block, each participant completes the three conditions in a counterbalanced order (e.g. a Latin square), with several targets per condition spanning a range of depths. **[TODO: fix trial counts and timings after piloting.]** Two considerations shape the procedure. First, because stereo and parallax can still produce a strong ceiling effect [6, 17], difficulty is raised deliberately: targets sit deep in the vasculature, behind other vessels so occlusion ordering is ambiguous (self-occlusion), and in visually crowded regions known to degrade depth perception [12]. Second, because free head movement recovers depth from parallax, viewing is constrained (e.g. a brief static-viewing window or system-driven rotation rather than unrestricted head movement) and a response deadline discourages extended viewpoint change. Head pose is logged throughout as a covariate. The exact depths, crowding, viewing-window duration, and deadline are fixed in a pilot that places performance off both floor and ceiling before the main run.

4.4 Error Metrics and Outcomes

For each placement, the 3D error vector between the marker and the ground-truth centreline node is decomposed — using the headset pose at placement — into a component along the view (camera) axis, the *depth* error, and an orthogonal *lateral* component. This separation is standard in 3D pointing because depth is the axis on which the cues act [16, 2], and only the depth component is diagnostic of a shader’s effect. The *signed* depth error is also recorded, to detect the systematic depth compression characteristic of head-mounted displays. The pre-registered primary outcome is the (unsigned) depth error; lateral error, signed depth error, and completion time are secondary.

4.5 Participants, Power, and Analysis

The perceptual sample will be drawn from a non-clinical population (e.g. computer-science and engineering students), on the rationale that the core depth judgement is a basic visual-system function and that prior work found similar novice/expert cue rankings for it [8]. Prior AR/VR experience and anatomy familiarity are collected as covariates, and an optional small expert arm allows a clinical-validity check. **[TODO: set target n from an a-priori power analysis once the pilot yields effect-size and variance estimates.]** The analysis is pre-registered: depth error is modelled with a linear mixed-effects model with shader condition as a fixed effect and a per-participant random intercept (random slope where supported), with target depth and logged covariates as controls. **[TODO: name the primary contrast(s) and the multiple-comparison correction.]** Because the literature predicts a possible null, the design is powered for the difficulty-manipulated contrast, and a null is treated as informative (Section 6).

Table 1. On-device rendering performance on Meta Quest 3. [TODO: fill from measurement.]

Mesh (triangles)	Foveation	Passthrough	Sustained FPS
[TODO]	off/on	on	[TODO]

5 Feasibility and Preliminary Results

Rendering performance. [TODO: report sustained stereo framerate for the vessel mesh over passthrough, with foveation on/off and against triangle count; confirm the headset display rate is held.]

Pilot (optional). [TODO: if run, a small ($n \approx 3\text{--}5$) pilot showing the apparatus yields a sensible depth-vs-lateral error decomposition and that the task is calibrated off floor/ceiling.]

6 Discussion, Limitations, and Conclusion

Interpreting the anticipated outcomes. Because passthrough MR does not trivially solve depth — the mesh competes with a textured background, the imagery is distorted and latent, and depth relative to the real scene is easily misjudged — depth-enhancement shaders have more room to help here than in immersive VR, where native cues dominate and shader differences wash out [6, 17]. A reliable improvement in the *depth* component of placement error would be directly useful for MR neuronavigation, whereas a null would itself be informative: even in this harder regime, and with a physical anchor supplying a strong external depth reference, added shader cues would not improve vessel depth judgement. The pre-registered analysis and the depth-versus-lateral decomposition make either outcome interpretable, and the difficulty calibration (Section 4.3) keeps the comparison sensitive.

Limitations. Several limitations bound the design. Rendering an opaque vessel *mesh* rather than the full volume is what makes the system run untethered and cleanly isolates the shader manipulation, but it abstracts away the surrounding tissue and the semi-transparent embedding that characterise the clinical view; the study therefore speaks to depth judgement of the vascular *surface* in situ rather than of vessels seen through translucent parenchyma. The subtlest remaining confound is the overlap between fog and any residual distance shading; we drive fog from an explicit depth function that blends toward a fixed reference colour (Section 4.2) and report its parameters in full. Passthrough adds its own confounds — geometric distortion, limited resolution, latency, and residual registration error — which we standardise across conditions and log. Because the mesh is anchored to the workspace rather than the patient, the study speaks

to MR *planning and inspection* rather than in-situ intraoperative navigation, and the anchor may aid localisation of the vasculature as a whole more than of an internal target vessel. Finally, a non-clinical sample bounds *perceptual*, not clinical, validity; because anatomically meaningful targets could let clinicians infer depth from anatomical priors [8], we use plausible but non-canonical target points and include an optional expert arm.

Conclusion and outlook. We have presented a standalone passthrough-MR system that renders a registered cerebral-vasculature mesh on the Meta Quest 3 and a rigorously controlled, pre-registered protocol for evaluating whether depth-enhancement shaders improve depth judgement in this setting, together with an on-device feasibility check. The immediate next step is to run the pre-registered study; beyond it, the clearest extensions are to make real-world occlusion a manipulated factor rather than a fixed baseline — the depth cue most specific to MR — and to move from a workspace anchor toward in-situ overlay on a patient phantom. **[TODO: close with the headline feasibility result once measured.]**

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